



PHIFEM: AN IMMERSED BOUNDARY FINITE ELEMENT METHOD FOR GEOMETRIES DEFINED BY A LEVEL-SET

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In this talk, we will present a fictitious domain finite element method called φ -FEM, well suited for elliptic problems posed in a domain given by a level-set function without requiring a mesh fitting the boundary. Unlike other recent fictitious domain-type methods (XFEM, Cut-FEM), our approach does not need any non-standard numerical integration, neither on the cut mesh elements nor on the actual boundary. We shall present the proofs of optimal convergence of our methods on the example of Poisson equation using Lagrange finite elements of any order. Furthermore, we highlight the flexibility and efficiency of our method on different problems (crack, interface, Stokes, heat equation,...). We will also give numerical tests illustrating the optimal convergence of our methods and discuss the conditioning of resulting linear systems and the robustness with respect to the geometry.

References

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